

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

22nd November 2019

Subject CM1A – Actuarial Mathematics (Paper A)

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** Differentiate between interest only loan and repayment loan. [2]
- Q. 2)** For any given age x , plot the “population mortality”, “ultimate mortality” and “select mortality”, varying by duration for an assurance contract and explain the shape of the plotted graph. [5]
- Q. 3)** i) Prove that
- $$(Ia)_{n|} = (\ddot{a}_{n|} - nv^n) / i$$
- Where i = effective rate of interest p.a. and n = term (2)
- ii) Mr Gary is working as an executive in a multi-national company. He is planning to start saving for his child’s higher education who is currently 8 years old. The expenses for higher education will be incurred after 10 years from now.
- The current cost of higher education is INR 30,00,000 which increases with an inflation of 6% p.a.
- Mr Gary plans to start the contribution by INR 2,00,000 p.a. immediately. From second year onwards, at the start of each year, he wishes to increase the previous year’s contribution by INR X till the 10th year. Assume that the interest earned on the savings is 10% p.a. effective, calculate how much should be the increase per year (i.e. X) so as to meet the fund requirement for higher education of the child at the end of tenth year. (4)
- [6]
- Q. 4)** i) What do you understand by the term *asset share*? Briefly describe how asset share could be used in determining terminal bonus to a policyholder? (2)
- ii) Everything else being equal, draw a comparison of premiums charged and expected benefits payable to policyholders when reversionary bonus is defined as: 3% simple bonus vs 3% compound bonus. (3)
- [5]
- Q. 5)** A motor insurer wishes to investigate whether gender of the policyholder plays any role in likelihood of having an accident and making a claim.
- Describe the steps that should be followed in the analysis of data for this investigation. [6]
- Q. 6)** An insurer sells 10-year term assurance policies with sum insured of INR 10 lakhs where benefit is payable at the end of the year of death and premiums are received annually in advance. 2,000 such policies were sold on 1st Jan, 2017 to lives aged 50 years.
- Basis:**
- Interest: 4 % p.a.
 - Mortality: AM92 Ultimate.
 - No expenses and other decrements to be assumed.
- i) Calculate the death strain at risk for each policy at the end of first year. (4)

- ii) There were 3 deaths in 2017 and 4 deaths in 2018 from this portfolio. Calculate the mortality profit or loss to the insurer at the end of year 2018. (3)
[7]

- Q. 7)** In evaluation of a long term benefit scheme for the employees in service, three decrements are considered – attrition rate, mortality rate and retirement rate. Attrition and mortality rates occur continuously over the year and retirement can occur only at age 60 years or 65 years.

Based on the past experience, you are given that the forces of decrement due to attrition rate and mortality rate are constant at 0.02 and 0.005 per annum respectively between 59.5 and 60.5 years. The retirement probability at exact age 60 years is 0.4.

If the workforce consists of 1,000 employees aged 59.5 years, calculate the number of employees withdrawing from the scheme on account of all decrements (attrition, death and retirement) between 59.5 and 60.5 years, rounded to nearest integer. [9]

- Q. 8)** i) Show that the duration of a perpetual bond with face value N delivering annual coupon c with yield to maturity (YTM) i is

$$= \frac{1+i}{i} \quad (2)$$

- ii) Let the duration of a portfolio P invested in 'n' bonds with weights w_i denominated in the same currency is the weighted average based on each bond's duration (D_i), i.e.

$$D_P = \sum_{i=1}^n w_i D_i$$

A portfolio manager implements a duration neutral strategy (i.e. $D_P = 0$) to protect his investment portfolio against variation in interest rates. The strategy consists of buying a bond (Bond A) and selling a bond (Bond B) both redeemable at par on maturity.

Features of the Bond A and Bond B are mentioned below:

Bond	Maturity	Coupon Rate p.a.	YTM p.a.	Quantity
A	18 yrs	9.5%	8%	100,000
B	20 yrs	10%	8%	?

Both coupon frequency & compounding frequency (YTM) are assumed to be semi-annual. YTM stands for yield to maturity. Calculate the following:

- a) Price and duration of Bond A. (3)
- b) Price and duration of Bond B. (3)
- c) Quantity of Bond B that the trader needs to sell so that the duration neutral strategy is implemented. (2)
[10]

- Q. 9)** A 5-year unit linked endowment assurance is issued to a male aged exactly 50. The cash flows in the non-unit fund at the end of the year is denoted by $(NUCF)_t$ where t takes values from 1 to 5 per policy which is in force at the start of year t are (-100, 10, 30, -40, 210). Independent mortality rate is AM92 Select and independent probability of withdrawal is 7% in the first three years and 1% in the last two years.
- i)** Calculate the reserves that are required at times $t = 1, \dots, 4$ in order to zeroise future negative cash-flows assuming the rate of accumulation to be 8% p.a. (5)
 - ii)** Calculate the net present value of the profits before and after zeroisation and state with reasons which of these would be greater. The risk discount rate is 9% p.a. (5)
 - iii)** Without carrying out any more calculations, explain the effect on the net present value if non-unit reserves earned interest at the rate of 9% p.a. (1)
- [11]**

- Q. 10)** A person is planning to purchase a car and is evaluating loan terms offered by two leading banks.

Bank A:

- One-time processing fee: 3% which will be charged upfront.
- Interest rate: 9% p.a. convertible monthly.
- On pre-payment of loan, minimum of X or Y will be levied towards foreclosure charges, where,
 $X = 5\%$ of the outstanding loan amount at the time of foreclosure
 $Y =$ the unpaid interest amount under the loan arrangement

Bank B:

- Processing fee: Nil.
- Interest rate: 10.5% p.a. convertible monthly.
- Nil pre-payment charges after 36 months, however 2% of the outstanding amount will be levied if paid earlier.

No part-payment of the loan is allowed in either of the banks.

Assume that the person is planning to take loan of INR 10,00,000 for a tenure of 8 years and is considering following options:

- a. Pre-payment at the end of 2nd year.
- b. Pre-payment at the end of 7th year.

Calculate the additional amount paid over and above the loan amount under both options. Hence state which bank the person should choose under each option.

[14]

Q. 11) A life insurance company sells a decreasing term assurance policy with following features:

- Original sum assured of INR 50,00,000 decreasing by INR 5,00,000 every year at the end of each policy year.
- Each contract has a policy term of 10 years.
- Policyholder is required to pay quarterly premiums for a period of 5 years only.
- Initial expenses related to each policy are 20% of the annual premium amount.
- Regular expenses are INR 500 p.a. & are assumed to be incurred annually at the start of each year from the second policy year.
- On the sale of each policy, the company pays an initial commission of 10% of annual premium; and
- Regular commission of 5% of premium income each time premium is received from the policyholder from second policy year onwards.

The insurance company has set the premium rate by using principle of equivalence.

For calculating the premium, the company assumes a mortality basis of 100% AM 92 table and interest rate of 6% p.a.

For calculating reserves, the company assumes same assumptions as above, except that all expenses (excluding commissions) are assumed to be higher by 10% and interest rate is assumed to be 4% p.a.

The insurance company has sold several such policies at the start of 2015 to male policyholders aged 30 years. For each such policy calculate the following:

- i)** Calculate the gross premium for product. (9)
- ii)** Assuming that 250 such policies are in-force at the beginning of 2019, calculate the actual profit over 2019 if actual mortality experience is 80% of expected; and investment income, expenses and other costs are as per the pricing basis. (16)

[25]
